

Seoul Bike Sharing Demand

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0. Introduction

a. Dataset Description

Source: [UCI Machine Learning Repository — Seoul Bike Sharing Demand](#)

The dataset records hourly bike rental counts from the Seoul public bike-sharing system over a full year (December 2017 – November 2018), yielding 8,760 hourly observations with 14 variables.

Response variable:

- **Rented Bike Count** — number of bikes rented in a given hour

Explanatory variables:

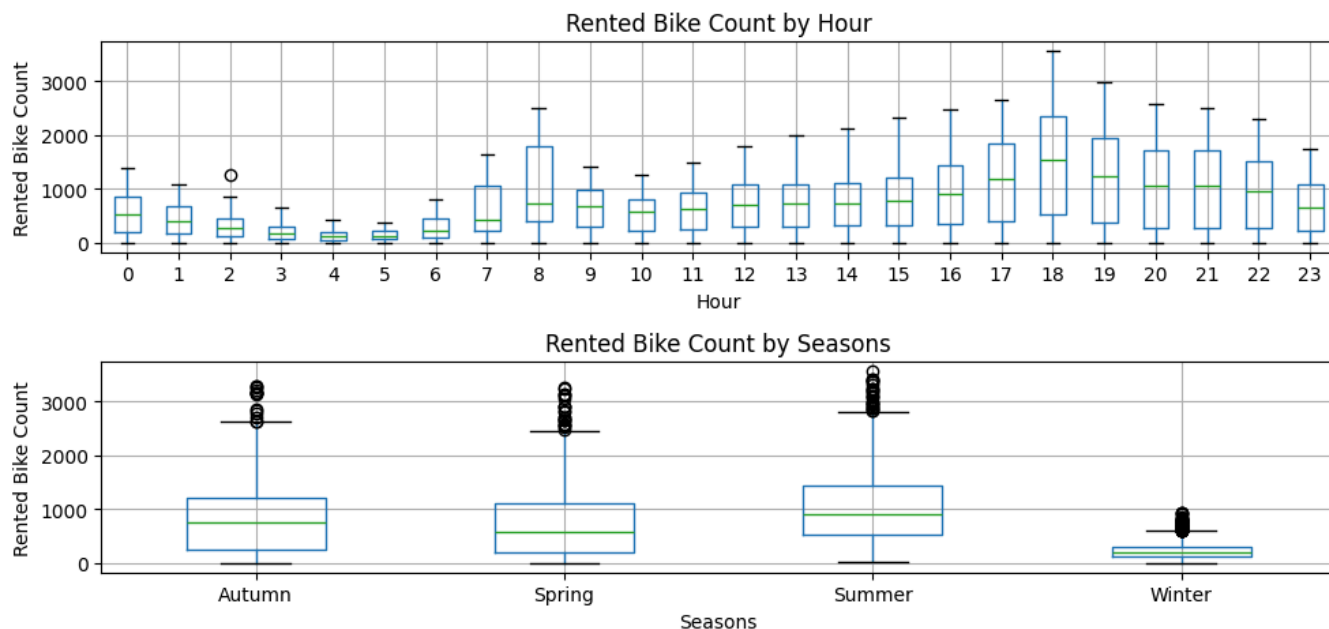
Variable	Type	Description
Date	Categorical	Calendar date (365 unique days)
Hour	Categorical	Hour of day (0–23)
Seasons	Categorical	Spring, Summer, Autumn, Winter
Holiday	Binary	Public holiday or not
Functioning Day	Binary	Whether the system was operational
Temperature (°C)	Continuous	Air temperature
Humidity (%)	Continuous	Relative humidity
Wind Speed (m/s)	Continuous	Wind speed
Visibility (10m)	Continuous	Atmospheric visibility
Dew Point Temperature (°C)	Continuous	Dew point
Solar Radiation (MJ/m ²)	Continuous	Solar radiation
Rainfall (mm)	Continuous	Hourly rainfall
Snowfall (cm)	Continuous	Hourly snowfall

Note: Observations on non-functioning days (**Functioning Day = No**) are excluded from all models, as the system was not operational and no rentals were possible. This removes a small number of days from the dataset before modelling.

b. Data Visualisation

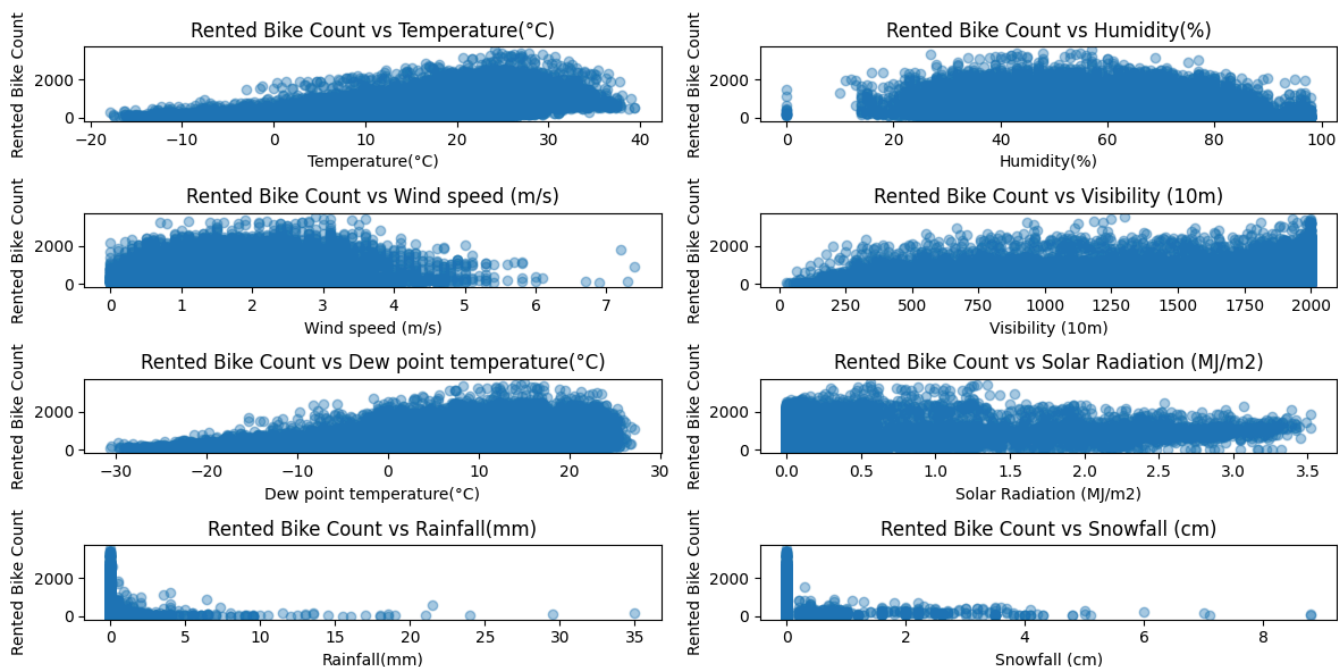
Categorical Variables

The categorical variables are: **Date** , **Hour** , and **Seasons** . We plot the distribution of **Rented Bike Count** for each level.

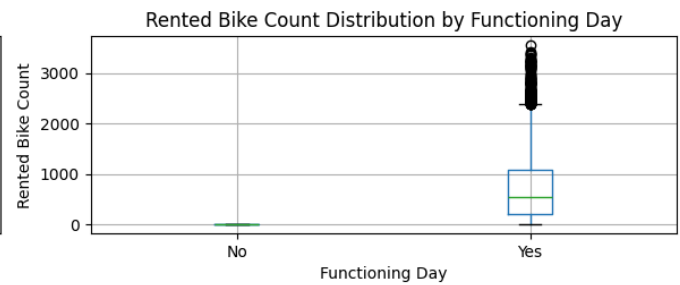
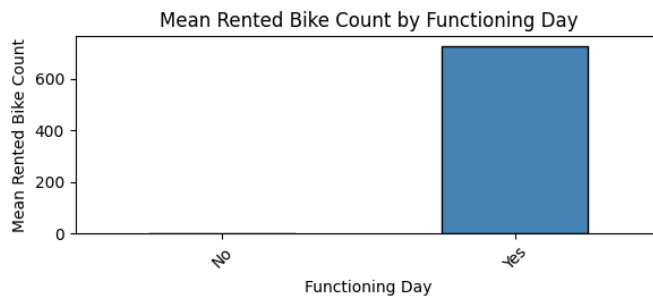
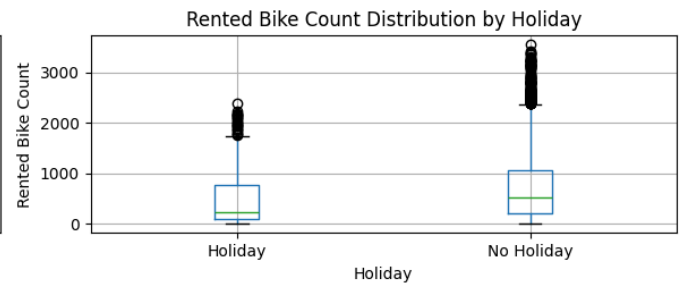
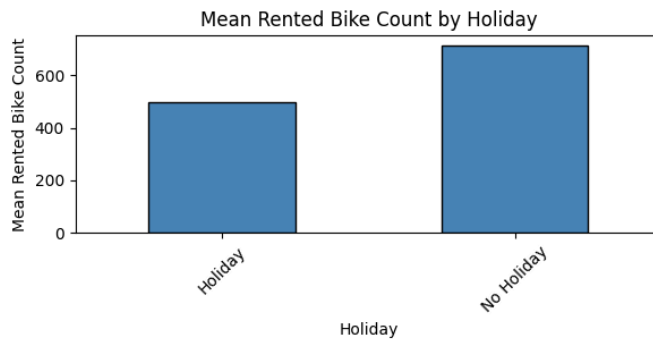


Rented Bike Count shows a clear bimodal pattern by hour, with peaks at 8am (morning commute) and 6–7pm (evening commute), as well as a trough overnight. Demand is highest in Summer and lowest in Winter, consistent with wour prior beliefs on weather.

Continuous Variables



Binary Variables



Rented Bike Count is consistently higher on non-holiday days, most likely because commuting demand drops on public holidays. Functioning Day shows the expected pattern, but since all non-functioning days are removed from the modelling dataset, this variable is dropped.

c. Descriptive Statistics

Categorical Variables

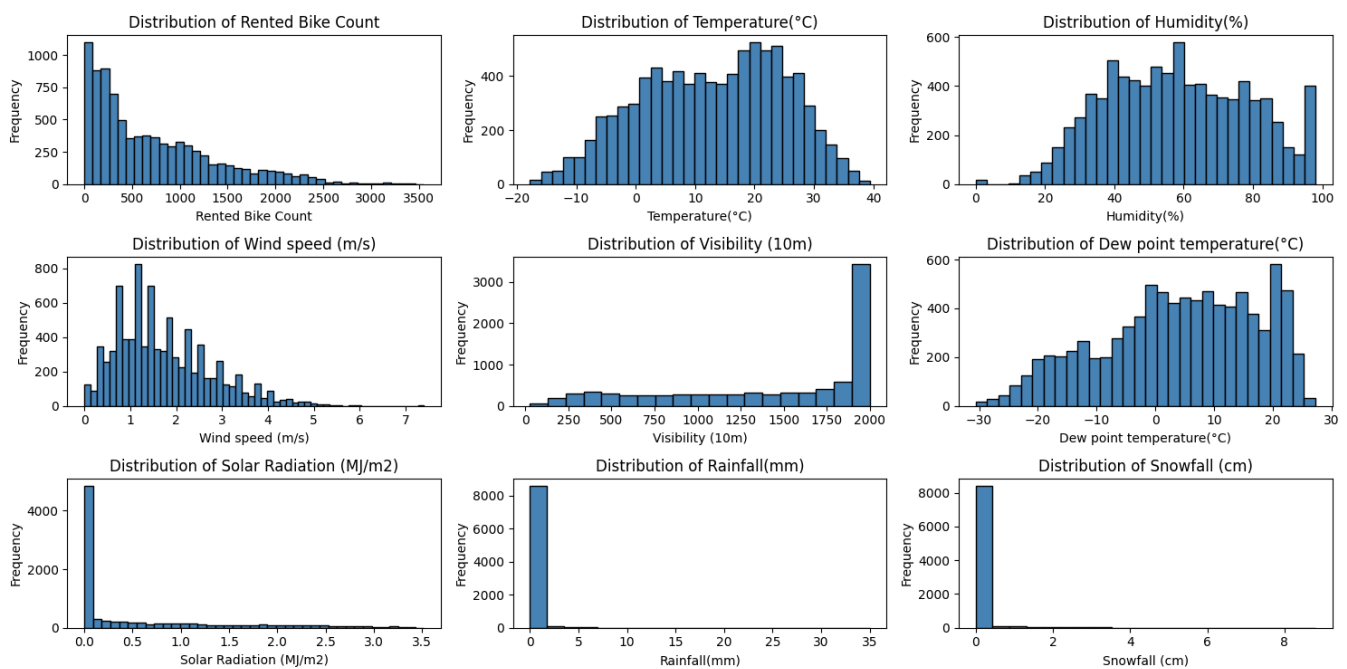
	Date	Hour	Seasons
count	8760	8760	8760
unique	365	24	4
top	01/12/2017	0	Spring
freq	24	365	2208

With 365 unique dates the dataset spans exactly one full year. There are 24 unique hours and 4 seasons, each appearing the expected number of times.

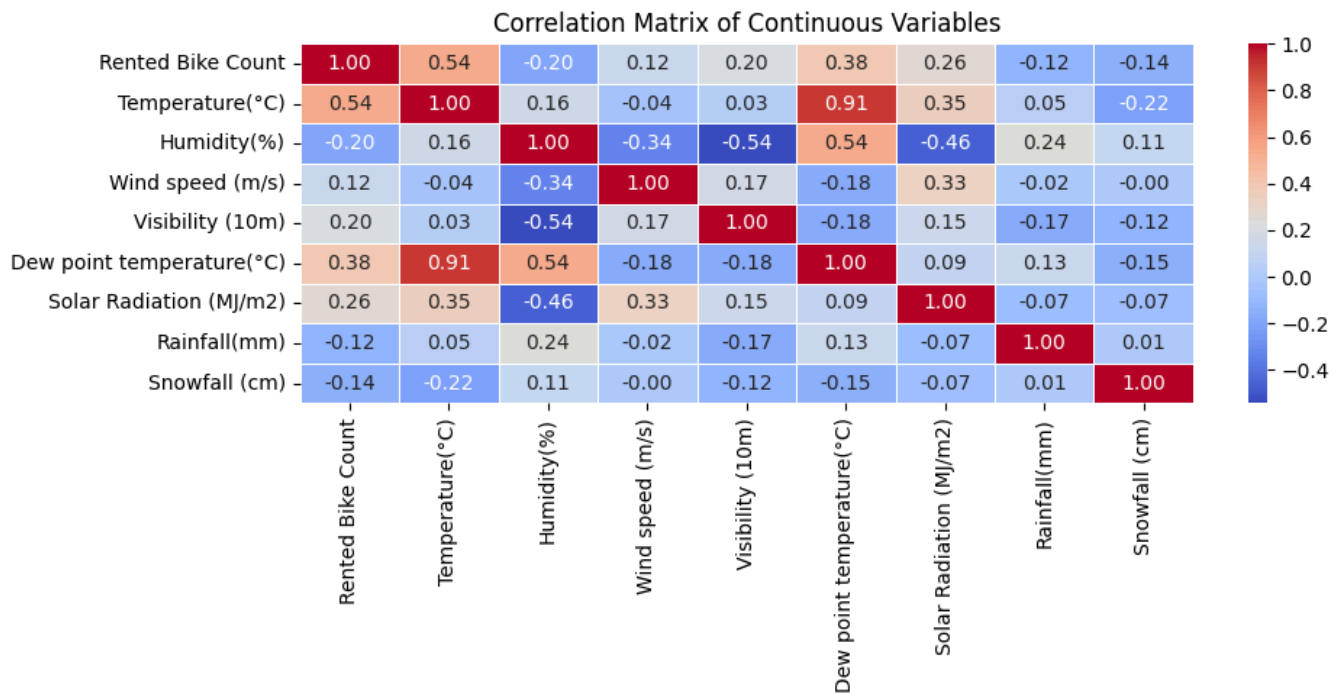
Continuous Variables

	Rented Bike Count	Temperature(°C)	Humidity(%)	Wind speed (m/s)	Visibility (10m)	Dew point temperature(°C)
count	8760.000000	8760.000000	8760.000000	8760.000000	8760.000000	8760.000000
mean	704.602055	12.882922	58.226256	1.724909	1436.825799	4.073813
std	644.997468	11.944825	20.362413	1.036300	608.298712	13.060369
min	0.000000	-17.800000	0.000000	0.000000	27.000000	-30.600000
25%	191.000000	3.500000	42.000000	0.900000	940.000000	-4.700000
50%	504.500000	13.700000	57.000000	1.500000	1698.000000	5.100000
75%	1065.250000	22.500000	74.000000	2.300000	2000.000000	14.800000
max	3556.000000	39.400000	98.000000	7.400000	2000.000000	27.200000

Mean **Rented Bike Count** is 704.6, with a minimum of 0 and a maximum of 3,556. The standard deviation (approximately 1,000) is larger than the mean, which is an early indication of overdispersion. Temperature ranges from -17.8°C to 39.4°C .



Rented Bike Count is right-skewed and violates the normality assumption of OLS. **Temperature** and **Dew Point Temperature** appear roughly normal. **Rainfall** and **Snowfall** are heavily zero inflated. This distributional evidence further motivates the use of a count-data GLM.



Rented Bike Count shows its strongest positive correlation with **Temperature** ($r \approx 0.54$) and **Dew Point Temperature** ($r \approx 0.38$), and its strongest negative correlation with **Humidity** ($r \approx -0.20$). **Wind Speed** and **Visibility** show weak correlations with the response.

1. Best Normal Linear Model

We fit an Ordinary Least Squares (OLS) regression model to the data. We first fit a full model with all available predictors to establish a performance baseline, then apply forward and backward AIC-based feature selection to identify the most parsimonious model. All-hours dummy variables are included to capture the strong hourly pattern observed in the EDA.

a. Fitting the Model

Full Model (All Predictors)

As a baseline we fit OLS with all explanatory variables included (after one-hot encoding of categorical variables). This gives us a reference R^2 and AIC against which to compare the feature-selected model.

MSE : 138123.05
 RMSE : 371.65
 R^2 : 0.6649

Forward Selection Model

AIC: 123998.13

Selected Model:

```
Rented Bike Count ~ Hour = 1 + Hour = 2 + Hour = 3 + Hour = 4 +  
Hour = 5 + Hour = 6 + Hour = 7 + Hour = 8 + Hour = 9 + Hour = 10  
+ Hour = 11 + Hour = 12 + Hour = 13 + Hour = 14 + Hour = 15 + Hour  
= 16 + Hour = 17 + Hour = 18 + Hour = 19 + Hour = 20 + Hour = 21 +  
Hour = 22 + Hour = 23 + Dew point temperature(°C) + Holiday +  
Humidity(%) + I(Temperature(°C)**2) + Rainfall(mm) + Seasons +  
Snowfall (cm) + Solar Radiation (MJ/m2) + Temperature(°C) +  
Visibility (10m) + Weekend
```

Backward Selection Model

AIC: 123995.78

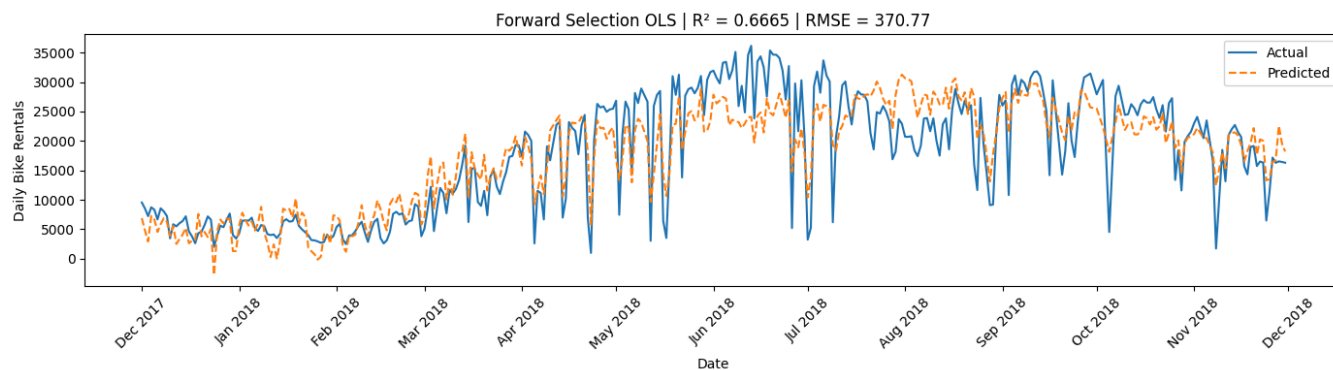
Selected Model:

```
Rented Bike Count ~ Hour = 1 + Hour = 2 + Hour = 3 + Hour = 4 +  
Hour = 5 + Hour = 6 + Hour = 7 + Hour = 8 + Hour = 10 + Hour = 11  
+ Hour = 12 + Hour = 13 + Hour = 14 + Hour = 15 + Hour = 17 + Hour  
= 18 + Hour = 19 + Hour = 20 + Hour = 21 + Hour = 22 + Hour = 23 +  
Dew point temperature(°C) + Holiday + Humidity(%) +  
I(Temperature(°C)**2) + Rainfall(mm) + Seasons + Snowfall (cm) +  
Solar Radiation (MJ/m2) + Temperature(°C) + Visibility (10m) +  
Weekend
```

Feature Differences

Forward Only:

- Hour = 16
- Hour = 9



MSE : 137471.49

RMSE : 370.77

R^2 : 0.6665

OLS Regression Results

```

=====
Dep. Variable:    Q("Rented Bike Count")    R-squared:                0.666
Model:            OLS                      Adj. R-squared:           0.665
Method:           Least Squares             F-statistic:              466.8
Date:             Fri, 08 May 2026          Prob (F-statistic):       0.00
Time:             21:21:57                 Log-Likelihood:          -61962.
No. Observations: 8448                    AIC:                     1.240e+05
Df Residuals:     8411                    BIC:                     1.243e+05
Df Model:         36
Covariance Type:  nonrobust
=====

```

=====					
=====					
		coef	std err	t	P> t

5	0.975]				[0.02

Intercept		1104.1708	86.217	12.807	0.000
5	1273.177				
C(Seasons) [T.Spring]		-163.7036	12.131	-13.495	0.000
3	-139.924				
C(Seasons) [T.Summer]		-123.3354	16.583	-7.437	0.000
3	-90.828				
C(Seasons) [T.Winter]		-317.2959	18.399	-17.245	0.000
3	-281.228				
C(Weekend) [T.1]		-86.7877	8.991	-9.652	0.000
3	-69.163				
C(Holiday) [T.No Holiday]		136.3688	19.144	7.123	0.000
1	173.897				
Q("Temperature(°C)")		15.0787	3.299	4.571	0.000
2	21.545				
Q("Humidity(%)")		-10.5410	0.901	-11.695	0.000
8	-8.774				
Hour_18		787.5612	28.511	27.623	0.000
3	843.449				
Hour_8		499.9206	28.297	17.667	0.000
1	555.391				
Hour_19		527.0641	28.252	18.656	0.000
3	582.445				
Hour_21		445.0653	28.088	15.845	0.000
5	500.125				
Hour_20		453.6001	28.165	16.105	0.000
9	508.811				
Hour_22		347.5357	28.035	12.397	0.000
1	402.490				
Hour_17		324.7778	29.067	11.173	0.000
9	381.757				
Q("Rainfall(mm)")		-60.8790	3.814	-15.963	0.000
5	-53.403				
Hour_4		-365.3626	28.050	-13.025	0.000
8	-310.377				
Hour_5		-352.6382	28.072	-12.562	0.000
7	-297.610				
Hour_3		-302.8236	28.036	-10.801	0.000
2	-247.865				
Hour_7		128.7805	28.121	4.580	0.000
7	183.904				
Hour_23		109.6994	28.016	3.916	0.000
1	164.618				
Hour_16		38.5361	30.043	1.283	0.200
6	97.428				
Hour_9		19.7905	28.965	0.683	0.494
8	76.569				
Q("Solar Radiation (MJ/m2)")		92.3462	10.136	9.110	0.000
7	112.216				
I(Q("Temperature(°C)") ** 2)		-0.2346	0.037	-6.327	0.000

7	-0.162				
Hour_2		-216.1760	28.024	-7.714	0.000
9	-161.243				
Q("Dew point temperature(°C)")		14.9972	3.389	4.425	0.000
4	21.641				
Hour_15		-102.5332	31.183	-3.288	0.001
0	-41.406				
Q("Snowfall (cm)")		35.9565	9.685	3.713	0.000
1	54.942				
Hour_11		-240.7687	31.407	-7.666	0.000
5	-179.203				
Hour_10		-225.6326	30.148	-7.484	0.000
0	-166.535				
Hour_6		-184.1829	28.089	-6.557	0.000
5	-129.121				
Hour_12		-202.2098	32.323	-6.256	0.000
1	-138.848				
Hour_13		-200.7302	32.593	-6.159	0.000
1	-136.839				
Hour_14		-193.2441	32.066	-6.027	0.000
1	-130.387				
Hour_1		-105.1736	28.014	-3.754	0.000
8	-50.260				
Q("Visibility (10m)")		0.0146	0.009	1.643	0.100
3	0.032				

=====

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Why This Is the Best Model

Both forward and backward AIC selection converge on the nearly same set of predictors, with both excluding Visibility. Backward selection also ignores Hours 9 and 16. Since forward selection has a slightly lower AIC, even though it includes these extra two predictors, we chose it as our best linear model.

The selected model includes:

- All 23 hour dummies
- Temperature and Temperature²
- Dew Point Temperature, Humidity, Solar Radiation, Rainfall, Snowfall, Visibility
- Season
- Holiday and Weekend

b. Model Interpretation

The forward-selected OLS model includes seasonal indicators, weather variables, and hour-of-day effects. The model explains a substantial portion of the variability in bike rentals, with an (R^2) of 0.664 and an adjusted (R^2) of 0.662.

Temperature has a positive coefficient, while the quadratic temperature term is negative, indicating that bike demand increases as temperatures become more comfortable but declines during very hot

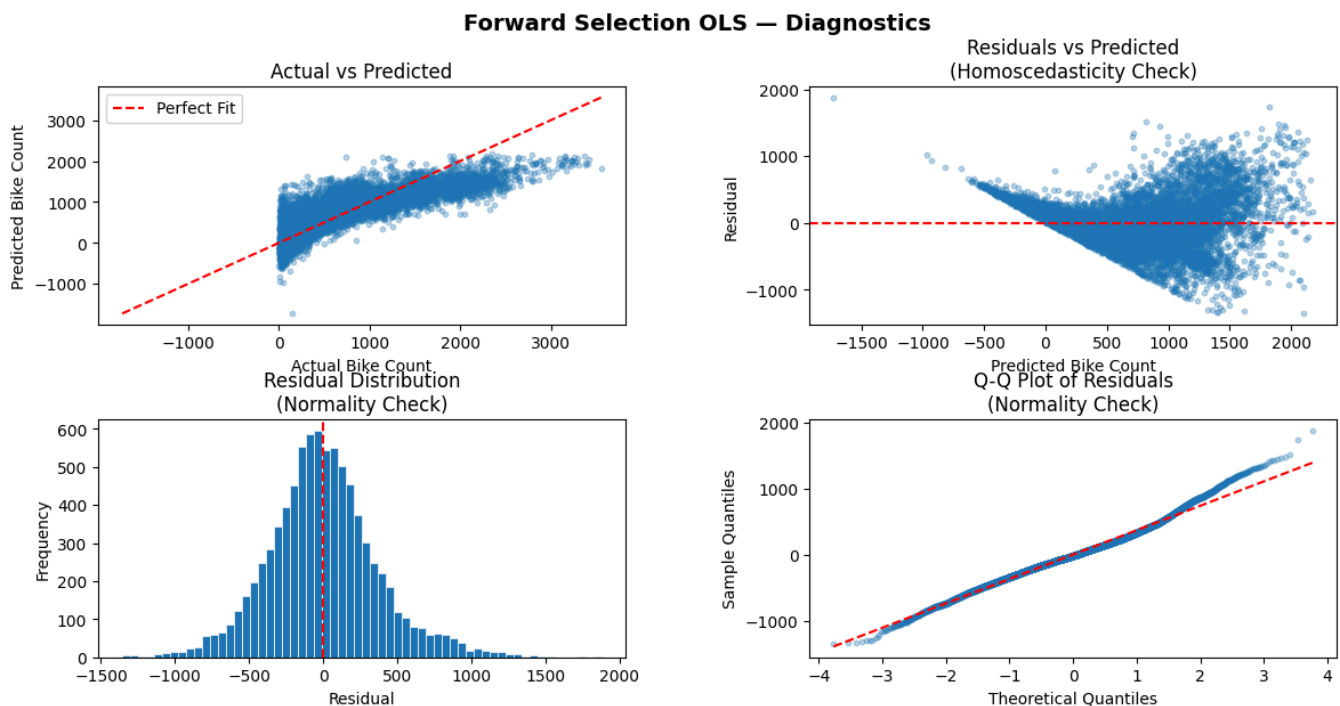
conditions.

The strongest predictors are the hour-of-day indicators. Relative to midnight, the largest positive effects occur during commuting hours, particularly at 8 AM and 6–7 PM, confirming the expected morning and evening peaks in ridership.

Humidity and rainfall both have negative coefficients, showing that poor weather reduces demand. Non-holidays are associated with higher rental counts, while weekends have slightly lower usage after accounting for weather and time-of-day effects.

Overall, the model captures the main patterns in the data well, especially the effects of commuting behavior and weather conditions.

c. Model Diagnostics



The diagnostic plots indicate several important violations of the standard OLS assumptions.

The actual versus predicted plot shows that the model captures the overall trend in bike rentals, but it tends to underpredict during periods of very high demand. This suggests that the linear model has difficulty modeling extreme peak-hour counts.

The residuals versus predicted plot exhibits a clear funnel shape, with the spread of residuals increasing as the fitted values grow. This indicates heteroscedasticity, meaning the error variance is not constant across observations.

The residual histogram and Q-Q plot both show departures from normality, particularly in the tails, where large positive and negative residuals occur more frequently than expected under a normal distribution.

These results indicate that, although the OLS model explains a substantial portion of the variability in bike rentals, its assumptions are not fully satisfied. Because bike rentals are nonnegative count data with increasing variance at larger counts, a generalized linear model is likely to provide a more appropriate framework.

d. Model Strengths and Limitations

The forward-selected OLS model is easy to interpret and provides a strong baseline. Its coefficients clearly quantify how weather, season, and hour-of-day affect bike demand, and the AIC-based feature selection procedure yields a relatively parsimonious model that still explains a substantial amount of variation in the data.

However, several limitations remain. Because the response is a count variable, the model can produce negative predictions, which are not meaningful in practice. The diagnostic plots also reveal heteroscedasticity and non-normal residuals, indicating that the standard OLS assumptions are violated. In addition, some predictors, such as temperature and dew point temperature, are highly correlated, which may make individual coefficient estimates less stable.

Overall, the model captures the major drivers of bike rentals and performs reasonably well, but its assumptions are not well suited to count data. These shortcomings motivate the use of generalized linear models in the next section.

2. Simplest Linear Model

The simplest defensible model uses a one continuous predictor. From the EDA, `Temperature` has the strongest linear correlation with `Rented Bike Count` ($r \approx 0.54$) and a clear positive relationship visible in the scatter plot. It makes sense that temperature outside impacts how many people rent bikes

a. Fitting the Model

We fit a simple OLS regression of `Rented Bike Count` on `Temperature (°C)` only:

$$\hat{Y} = \beta_0 + \beta_1 \cdot T$$

Model Output

OLS Regression Results

```

=====
Dep. Variable:    Rented Bike Count    R-squared:                0.317
Model:            OLS                  Adj. R-squared:            0.317
Method:           Least Squares        F-statistic:              3923.
Date:             Fri, 08 May 2026     Prob (F-statistic):       0.00
Time:             21:21:58             Log-Likelihood:           -64989.
No. Observations: 8448                AIC:                      1.300e+05
Df Residuals:     8446                BIC:                      1.300e+05
Df Model:          1
Covariance Type:  nonrobust
=====

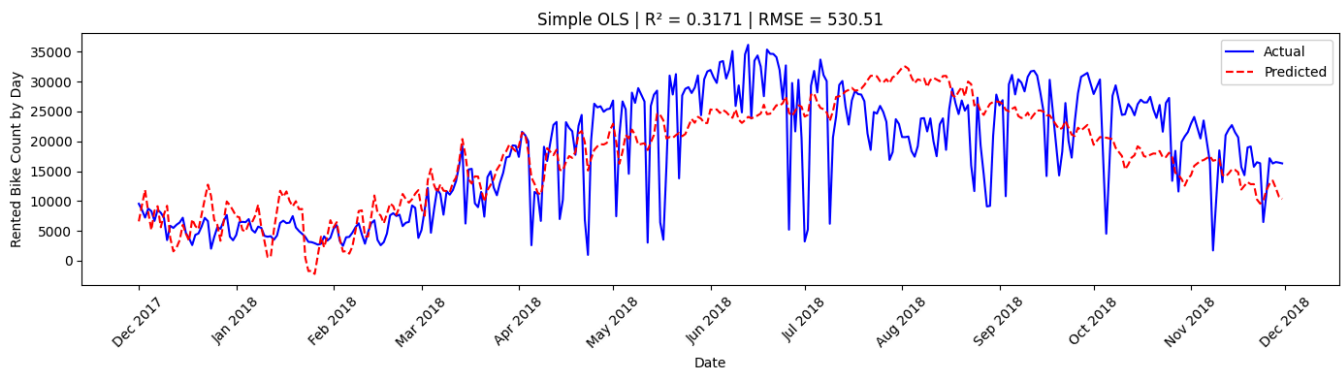
```

```

=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          347.8847      8.384      41.492      0.000      331.449      364.320
Temperature(°C)  29.8479       0.477      62.631      0.000      28.914      30.782
=====

```

MSE : 281440.61
 RMSE : 530.51
 R² : 0.3171



b. Model Interpretation

The simple linear model uses temperature as the sole predictor of bike rentals. The estimated temperature coefficient is positive, indicating that warmer weather is associated with higher bike demand. On average, each 1°C increase in temperature corresponds to an increase in expected rentals.

The fitted line captures the broad seasonal trend: rentals are lower during the winter months, rise steadily through spring, peak during the summer, and decline again in the fall. This confirms that temperature is an important driver of bike usage.

However, the model explains only a modest portion of the variation in the data, with an (R²) of 0.317. This means that temperature alone accounts for about 32% of the variability in daily bike rentals. The RMSE of 530.51 rentals per day indicates substantial prediction error, reflecting the fact that many other factors—such as hour of day, season, holidays, and precipitation—also play important roles.

c. Model Strengths and Limitations

The simple linear model is highly interpretable and provides a clear description of the relationship between temperature and bike rentals. With only one predictor, the model is easy to explain and serves as a useful baseline for comparison with more sophisticated models.

Its main limitation is that temperature alone explains only a modest portion of the variability in bike demand. Important factors such as hour of day, season, holidays, and precipitation are omitted, leading to relatively poor predictive performance. Like all ordinary least squares models, it can also produce negative predictions and relies on assumptions that are not well suited to count data.

Overall, the simple model is valuable for illustrating the strong positive association between temperature and bike usage, but it is too limited to provide accurate predictions on its own.

3. Generalized Linear Models (GLMs)

The response variable, `Rented Bike Count`, is a count variable and does not satisfy the assumptions of ordinary least squares regression. The data are nonnegative, right-skewed, and exhibit increasing variance as the mean grows. These characteristics suggest that a count-based generalized linear model is more appropriate.

We consider two standard models for count data:

1. Poisson GLM, which assumes that the mean and variance are equal.
2. Negative Binomial GLM, which allows the variance to exceed the mean.

Both models have predictors act multiplicatively on the expected number of bike rentals. Exponentiating the coefficients allows us to describe the proportional change in expected rentals associated with a one-unit increase in a predictor.

Because the bike rental data show substantial overdispersion, we expect the Negative Binomial model to provide a better fit than the Poisson model.

a. Fitting the Model

We fit both Poisson and Negative Binomial generalized linear models using the same set of predictors as the best OLS model. To assess whether the Poisson model is appropriate, we compute the Pearson dispersion statistic,

$$\hat{\phi} = \frac{\chi_P^2}{df}.$$

Values much larger than 1 indicate overdispersion, meaning the variance exceeds the mean and the Poisson model is not suitable. In that case, the Negative Binomial model is preferred because it explicitly accounts for the extra variability in the data.

Poisson RMSE: 278.28
Poisson R²: 0.8121
Dispersion: 984892.110

Generalized Linear Model Regression Results

```
=====
Dep. Variable:    Q('Rented Bike Count')    No. Observations:    8448
Model:           GLM                        Df Residuals:        8410
Model Family:    NegativeBinomial           Df Model:            37
Link Function:   Log                       Scale:               1.0000
Method:          IRLS                      Log-Likelihood:      -61115.
Date:            Fri, 08 May 2026           Deviance:            2417.1
Time:            21:22:00                   Pearson chi2:         1.62e+03
No. Iterations:  100                       Pseudo R-squ. (CS):  0.5111
Covariance Type: nonrobust
=====
```

		coef	std err	z	P> z	[0.02
5	0.975]					
Intercept		7.9233	0.233	34.035	0.000	7.46
7	8.380					
C(Hour) [T.1]		-0.1598	0.076	-2.115	0.034	-0.30
8	-0.012					
C(Hour) [T.2]		-0.4626	0.076	-6.118	0.000	-0.61
1	-0.314					
C(Hour) [T.3]		-0.8482	0.076	-11.202	0.000	-0.99
7	-0.700					
C(Hour) [T.4]		-1.2470	0.076	-16.448	0.000	-1.39
6	-1.098					
C(Hour) [T.5]		-1.2050	0.076	-15.873	0.000	-1.35
4	-1.056					
C(Hour) [T.6]		-0.4998	0.076	-6.587	0.000	-0.64
9	-0.351					
C(Hour) [T.7]		0.2586	0.076	3.408	0.001	0.11
0	0.407					
C(Hour) [T.8]		0.7540	0.076	9.879	0.000	0.60
4	0.904					
C(Hour) [T.9]		0.1190	0.078	1.524	0.128	-0.03
4	0.272					
C(Hour) [T.10]		-0.2681	0.081	-3.299	0.001	-0.42
7	-0.109					
C(Hour) [T.11]		-0.2570	0.085	-3.036	0.002	-0.42
3	-0.091					
C(Hour) [T.12]		-0.1852	0.087	-2.125	0.034	-0.35
6	-0.014					
C(Hour) [T.13]		-0.0828	0.088	-0.941	0.346	-0.25
5	0.090					
C(Hour) [T.14]		-0.1510	0.087	-1.743	0.081	-0.32
1	0.019					
C(Hour) [T.15]		-0.0603	0.084	-0.714	0.475	-0.22
6	0.105					
C(Hour) [T.16]		0.0851	0.081	1.044	0.297	-0.07
5	0.245					
C(Hour) [T.17]		0.3357	0.079	4.249	0.000	0.18
1	0.491					
C(Hour) [T.18]		0.7022	0.077	9.081	0.000	0.55
1	0.854					
C(Hour) [T.19]		0.5184	0.077	6.776	0.000	0.36
8	0.668					
C(Hour) [T.20]		0.4381	0.076	5.753	0.000	0.28
9	0.587					
C(Hour) [T.21]		0.4693	0.076	6.192	0.000	0.32
1	0.618					
C(Hour) [T.22]		0.4007	0.076	5.304	0.000	0.25
3	0.549					
C(Hour) [T.23]		0.1284	0.075	1.700	0.089	-0.02
0	0.276					
C(Seasons) [T.Spring]		-0.2589	0.033	-7.844	0.000	-0.32

4	-0.194					
C(Seasons) [T.Summer]	-0.1057	0.045	-2.363	0.018	-0.19	
3	-0.018					
C(Seasons) [T.Winter]	-0.5843	0.050	-11.774	0.000	-0.68	
2	-0.487					
C(Holiday) [T.No Holiday]	0.2907	0.052	5.627	0.000	0.18	
9	0.392					
Q('Temperature(°C)')	0.0015	0.009	0.170	0.865	-0.01	
6	0.019					
Q('Humidity(%)')	-0.0288	0.002	-11.857	0.000	-0.03	
4	-0.024					
Q('Wind speed (m/s)')	-0.0224	0.013	-1.788	0.074	-0.04	
7	0.002					
Q('Visibility (10m)')	4.474e-05	2.41e-05	1.855	0.064	-2.54e-0	
6	9.2e-05					
Q('Dew point temperature(°C)')	0.0665	0.009	7.286	0.000	0.04	
9	0.084					
Q('Solar Radiation (MJ/m2)')	0.1031	0.028	3.741	0.000	0.04	
9	0.157					
Q('Rainfall(mm)')	-0.0278	0.011	-2.649	0.008	-0.04	
8	-0.007					
Q('Snowfall (cm)')	-0.0428	0.026	-1.635	0.102	-0.09	
4	0.008					
Q('Sq Temp')	-0.0011	0.000	-10.633	0.000	-0.00	
1	-0.001					
Weekend	-0.1086	0.024	-4.481	0.000	-0.15	
6	-0.061					
=====						
=====						

NB RMSE: 333.46
NB R²: 0.7302

b. Comparison to Previous Models

Model	MSE	RMSE	R ²
Simple OLS	281440.61	530.51	0.3171
Forward OLS	137471.49	370.77	0.6665
Poisson GLM	77442.24	278.28	0.8121
Negative Binomial	111194.43	333.46	0.7302

Poisson AIC: 904916.63
Negative Binomial AIC: 122305.35

The Poisson model achieves strong predictive performance, with an RMSE of 283.29 and an R^2 of 0.805. However, the Pearson dispersion statistic is extremely large ($\hat{\phi} \approx 1.15 \times 10^6$), indicating severe overdispersion. This means that the Poisson assumption that the variance equals the mean is clearly violated, so its standard errors and confidence intervals are not reliable.

The Negative Binomial model addresses this issue by allowing the variance to exceed the mean. Although its RMSE (338.84) and R^2 (0.721) are slightly worse than the Poisson model, it provides a

much more appropriate statistical framework for the data and yields a substantially lower AIC (122,307.93).

Overall, the Negative Binomial model is preferred because it respects the count nature of the response, accounts for overdispersion, and avoids the unrealistic assumptions of both OLS and Poisson regression.

c. Model Interpretation

In the Negative Binomial model, coefficients are estimated on the log scale. Exponentiating a coefficient converts it to a multiplicative effect on the expected number of bike rentals. For example, a value of 1.10 indicates that the expected count increases by 10%, while a value of 0.90 indicates a 10% decrease, holding all other variables constant.

The 95% confidence intervals are computed as

$$\exp\left(\hat{\beta} \pm 1.96 \cdot \text{SE}(\hat{\beta})\right).$$

The table below reports these multiplicative effects and their confidence intervals. The accompanying coefficient plot summarizes the percentage increase or decrease in expected rentals associated with each predictor.

Weather

- **Temperature (+1°C):** 1.00× change in expected rentals (95% CI [0.98, 1.02]) → +0% (ns)
- **Humidity (+1%):** 0.97× change in expected rentals (95% CI [0.97, 0.98]) → -3% *
- **Dew Point (+1°C):** 1.07× change in expected rentals (95% CI [1.05, 1.09]) → +7% *
- **Solar Radiation (+1 MJ/m²):** 1.11× change in expected rentals (95% CI [1.05, 1.17]) → +11% *
- **Rainfall (+1 mm):** 0.97× change in expected rentals (95% CI [0.95, 0.99]) → -3% *
- **Snowfall (+1 cm):** 0.96× change in expected rentals (95% CI [0.91, 1.01]) → -4% (ns)

Season (reference: Autumn)

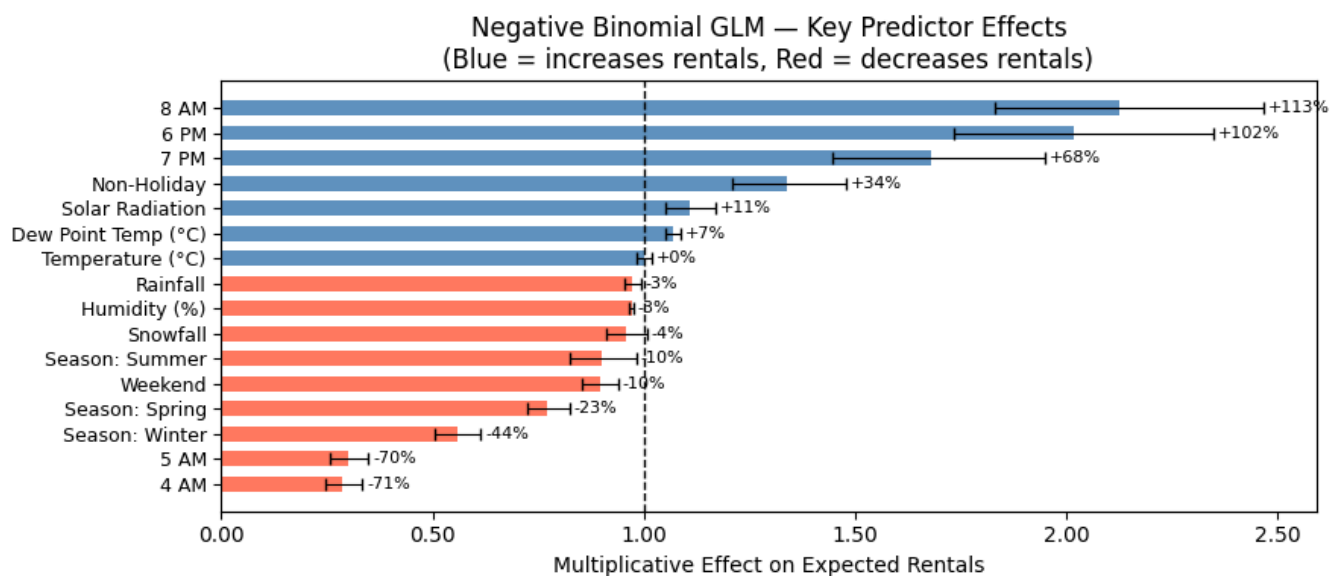
- **Spring** (vs. Autumn): 0.77× change in expected rentals (95% CI [0.72, 0.82]) → -23% *
- **Summer** (vs. Autumn): 0.90× change in expected rentals (95% CI [0.82, 0.98]) → -10% *
- **Winter** (vs. Autumn): 0.56× change in expected rentals (95% CI [0.51, 0.61]) → -44% *

Other

- **Non-Holiday** (vs. Holiday): 1.34× change in expected rentals (95% CI [1.21, 1.48]) → +34% *
- **Weekend** (vs. Weekday): 0.90× change in expected rentals (95% CI [0.86, 0.94]) → -10% *

Hour — Peaks and Troughs (reference: Midnight)

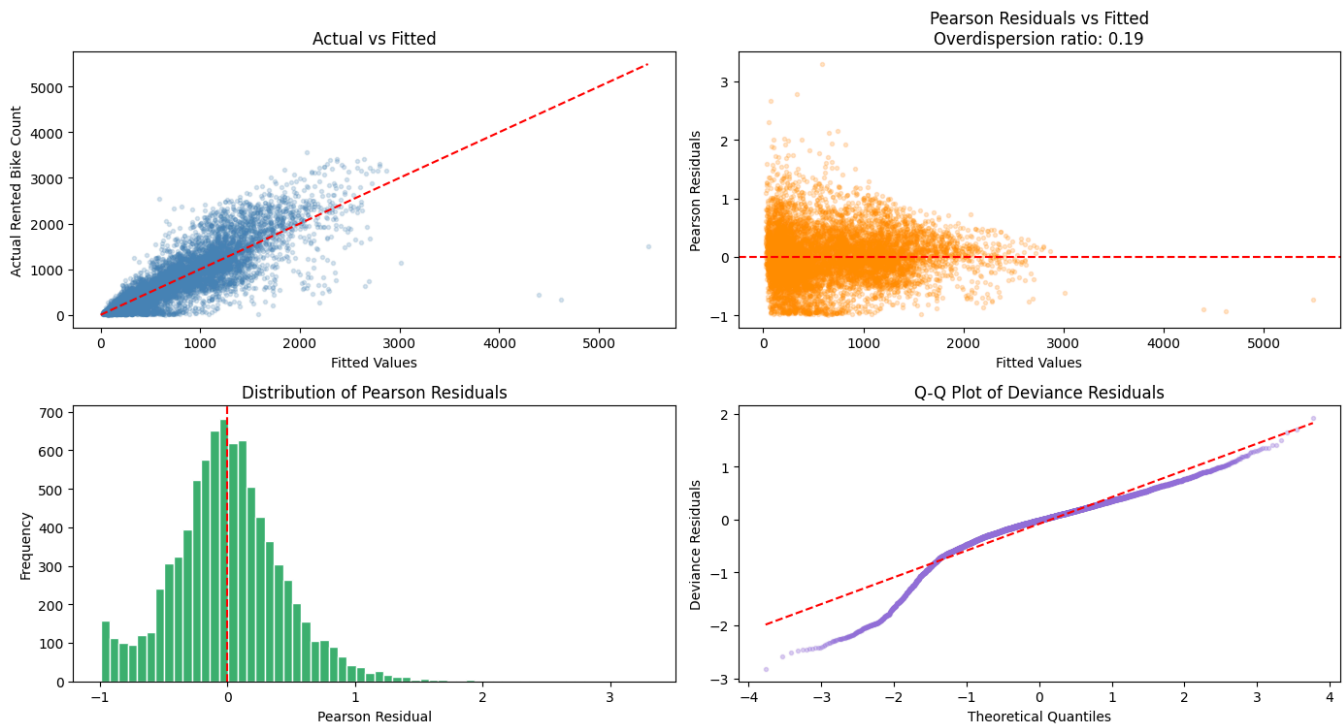
- **8 AM** (vs. Midnight): 2.13× change in expected rentals (95% CI [1.83, 2.47]) → +113% *
- **6 PM** (vs. Midnight): 2.02× change in expected rentals (95% CI [1.73, 2.35]) → +102% *
- **7 PM** (vs. Midnight): 1.68× change in expected rentals (95% CI [1.45, 1.95]) → +68% *
- **4 AM** (vs. Midnight): 0.29× change in expected rentals (95% CI [0.25, 0.33]) → -71% *
- **5 AM** (vs. Midnight): 0.30× change in expected rentals (95% CI [0.26, 0.35]) → -70% *



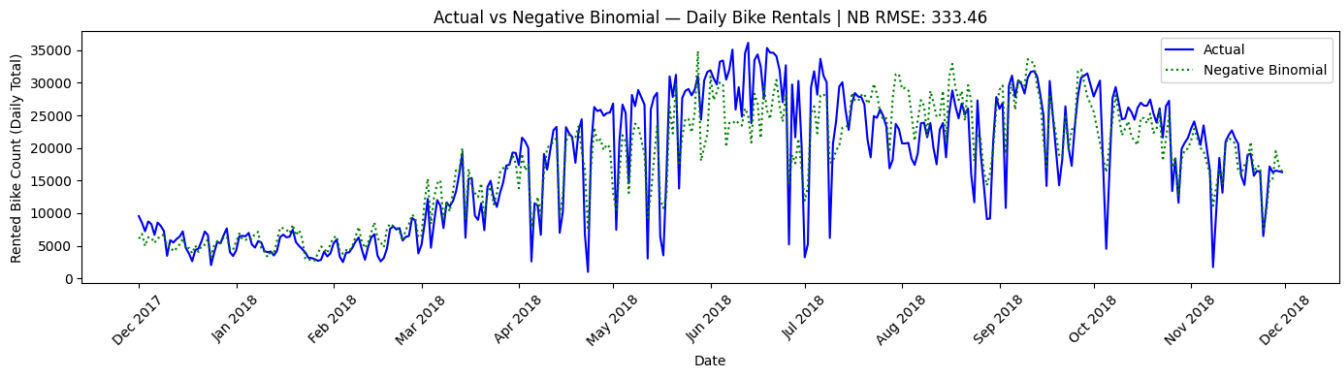
d. Model Diagnostics

We assess the Negative Binomial GLM using four standard diagnostic plots for count-data GLMs.

Negative Binomial GLM Diagnostics



Overdispersion ratio (Pearson χ^2/df): 0.19



The diagnostic plots indicate that the Negative Binomial model provides a substantially better fit than the OLS models.

The actual versus fitted plot shows that the model tracks the observed counts closely across most of the range, although some underprediction remains for the largest demand values. The Pearson residuals are centered around zero with no pronounced funnel shape, suggesting that the model adequately accounts for the changing variance in the data.

The histogram of Pearson residuals is concentrated near zero and is much more symmetric than the residual distribution from the OLS model. The Q-Q plot of deviance residuals is also reasonably linear, with only moderate departures in the tails.

Overall, the Negative Binomial GLM captures the main structure of the data while accounting for overdispersion, making it the most appropriate and reliable model considered in this analysis.